

# Are Long-Running Agents Making Progress?

## A Longitudinal Study of Artifact Progress, Repeated Mutation, and Continuity

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### Abstract

Long-running agents can modify a workspace for days across many session boundaries. When they stop, action counts, commits, and final artifacts do not reveal whether activity accumulated into durable and validated progress or repeatedly overwrote, abandoned, and rediscovered the same work. Existing trajectory studies characterize actions within benchmark attempts, while studies of long-lived agents focus on task outcomes or internal memory degradation. We instead make the persistent workspace the longitudinal unit of analysis. We introduce a source-linked projection that connects native Claude, Codex, and Gemini actions to stable artifact identities, lifecycle effects, event time, and session boundaries without introducing semantic labels or aligning the timeline to Git commits. We use it to study six real software and autonomous- research projects through introduced-artifact persistence, later reuse, validation association, repeated mutation, workspace-activity migration, and cross-session continuity. Under the admitted local projection, across 551 native root sessions and 181,303 Tool actions, 89.3–97.0% of eligible mutations are later revisited, yet action volume only weakly orders the cross-project reuse relation ( $\rho = 0.20$ ). Repeat-observed episodes account for 74.5–90.8% of mutation episodes, with 42.0–86.7% of episodes concentrated in the most-mutated 10% of identities. Source coverage is itself consequential: recognized successful validation appears in all six cases after projection repair, and named Skills show no supported repeatable separation in the only eligible comparison. In 320 stratum-specific public task/topic selections, path-local transitions and 2–3-intervening-call module returns recur, but persistent artifact lineage and cross-session re-grounding remain unobservable. Finally, a source-direct 120-question conformance experiment found the frozen implementation answering only 32/60 artifact-linked and cross-session facts correctly (2026-07-23). Root-cause repair of session joins, failed-call effects, shell path extraction, and event workdir handling brings the current revision to 60/60 on the corrected benchmark—a repair-corpus conformance result, not a general exact-fact capability claim.

### Introduction

Long-running agents increasingly receive a broad goal, work in one repository for days, and return a workspace containing

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code, prose, tests, and experimental output. Tool calls and commits show activity, but not whether that activity accumulated into durable, reused, and validation-associated progress. Final state omits reads, failed validations, transient artifacts, overwritten attempts, and the work of re-establishing context after a session boundary. Raw logs preserve these events but not the cross-session artifact lineage needed to connect them.

Trajectory studies already characterize action sequences, validation behavior, and destructive revision in benchmark attempts (Bouzenia and Pradel 2025; Mehtiyev and Assunção 2026; Kim et al. 2026). ProcGrep represents such traces as action procedures (Oderinwale 2026); persistent workspaces and multi-session reliability are also established settings (Zhou et al. 2026; Zhu et al. 2026). Context management can rapidly weaken earlier plans (Mehta and Datta 2026). The remaining gap is longitudinal and artifact-centered: a project persists while its sessions end, and its files acquire histories that attempt-level action summaries do not express.

We therefore keep three observable dimensions separate. *Introduced-artifact persistence* asks whether a confirmed introduction survives in the final workspace; *reuse* asks whether later source-linked work returns to the same artifact lineage; and *validation association* asks whether an adapter-recognized successful check occurs before the artifact’s next mutation or deletion. Their conjunction is descriptive shorthand, not a weighted quality score. We likewise treat repeated edits, unvisited artifacts, and validation gaps as process facts rather than intent, waste, or failure labels.

Our source-linked projection retains native Claude, Codex, and Gemini Tool actions, resolves qualified lifecycle effects, preserves explicit rename and session lineage, and orders events by Agent action time. Git contributes final tracked state but never defines the timeline. The same source facts support the quantitative analyses and auxiliary AGENT NEBULA replay; visual coordinates never enter the measurements. Independent source-direct checkers test the projection rather than accepting its rows as truth.

Across 551 native root sessions and 181,303 Tool actions in six projects, eligible-mutation reuse is 89.3–97.0% but action volume only weakly orders the six-case reuse relation ( $\rho = 0.20$ ). Repeat-observed mutations are common and concentrated; recognized successful validation covers all 6/6 cases after repair. The only eligible Skill comparison shows no repeatable fingerprint, while public traces reproduce path

locality and 2–3-call module returns but not persistent lineage. Finally, source-direct conformance moves from a frozen 32/60 artifact-linked and cross-session facts to 60/60 after root-cause repair against the corrected benchmark.

This paper contributes (1) a workspace-centered longitudinal characterization of artifact progress and continuity; (2) a deterministic, source-linked protocol with explicit measured, coverage-only, disagreement, and N/A states; and (3) a conformance gate whose frozen 32/60 artifact-linked and cross-session result motivated root-cause repair to 60/60 against a corrected oracle on this corpus. The supplement contains complete definitions, per-project results, all moved figures, and the extended validity analysis.

## Workspace-Centered Measurement

**Source universe and order.** A workspace  $W$  persists while native root sessions  $S = (s_1, \dots, s_m)$  issue Tool actions  $A = (a_1, \dots, a_n)$ . Sessions partition model context but do not reset  $W$ . We admit a complete session when its project directory, worktree, or Git remote identifies a selected repository, and retain every Tool action in that session, including failed, no-path, and search actions. Native timestamp and stable source ID order the trace; global path-only matches remain a separate sensitivity source.

**Action and effect projection.** For each action we retain

$$a_i = (t_i, id_i, s_i, v_i, u_i, e_i, q_i), \quad E_i = \{(w, f, o, f')\}, \quad (1)$$

where vendor  $v_i$ , Tool/category  $u_i$ , adapter-derived command effect  $e_i$ , and status  $q_i$  accompany zero or more effects on worktree  $w$  and relative artifact  $f$ . Operation  $o$  is read, write, create, rename, or delete;  $f'$  exists only for an observed rename. Failed calls remain actions but emit no confirmed-success effect. Missing fields and unresolved paths remain coverage gaps.

**Artifact lineage and outcomes.** Identity is a hashed worktree and relative path. Tool-level workdir overrides session cwd; an explicit same-worktree rename preserves lineage and birth state; delete followed by create starts a new identity. Only a confirmed-success create is an eligible introduction. For every confirmed non-delete mutation, we locate later access and the next recognized successful test, check, or build before another same-artifact mutation or deletion. Supersession is a competing outcome; only observation end is right-censored. Final existence and tracked state are independently queried in the associated worktree. A recognized success establishes temporal association, not mutation-level coverage or correctness.

**Observable progress without semantic gold.** The projection supports three source-verifiable outcomes. Persistence uses final existence or tracked state only for artifacts observably introduced during the trace; writes to pre-existing files have unknown content durability without independent diffs or snapshots. Reuse requires a later access to the same lineage and retains event, time, and session distance. Validation association measures distance from a mutation to the next recognized successful validation before supersession. Directory arguments are weak scope evidence rather than fabricated

reads of all descendants, and unknown status is neither success nor failure. Complete distance curves or declared sensitivity ranges replace a fixed attention window. A never-revisited file is an observed orphan, not automatically useless; repeated edits remain uninterpreted without task evidence.

**Protocol and implementation.** Each repository is one complete case. Cross-case summaries state project weighting and never treat actions as independent projects; blocked bootstrap or permutation tests use session or project blocks only when exchangeability is defensible. Artifact class is a deterministic path/extension mapping and top-level directories are the default module unit. The Rust implementation extends an existing repository projection, reuses native `agent-session` parsing, and exports source-linked JSON/CSV rows. Structured events, rather than `AGENT NEBULA` layout coordinates, drive every statistic. Independent checkers reparse native sources for the conformance, Skill, and public-trace analyses.

**Capability protocol.** The separate measurement question holds source membership and cutoff state fixed, stratifies facts as action-only, artifact-linked, cross-session, or final-state, and requires every answer to cite source evidence or abstain. Final State and Counts are lower-information controls, pinned ProcGrep is the strongest action-procedure baseline, and a bounded Raw reader was planned as a same-source reconstruction condition. No method labels its own oracle. Deterministic conditions complete 480 rows; the Raw branch remains N/A after its preflight boundary stop, so the planned 840-row integrated comparison is not represented as complete.

## Empirical Study

**Research questions.** **RQ1** asks how introduced and repeatedly modified artifacts persist, receive later reuse, concentrate work, become dormant, and return over Agent action time. **RQ2** asks how successful and failed validations interleave with mutation bursts, backlog, and subsequent workspace changes. **RQ3** asks how path-resolved activity is allocated among artifact types and how artifact/module hotspots migrate and receive returns. **RQ4** asks what re-grounding and prior-artifact/module continuity is observable between adjacent non-overlapping native-root components. **RQ5** asks whether workspace-action composition attributed to named Skills is repeatable across independent roots and what follows explicit instruction-file access. **RQ6** asks which compatible within-attempt relations recur in public coding and scientific-process traces and which longitudinal claims remain N/A.

**Cases and inclusion.** We fix six local open-source repositories before computing process rates: AgentSight and ActPlane (systems software and research), bpf-developer-tutorial (tutorial/code/documentation), eunomia.dev (content/documentation/software), agentskill-observability-paper (paper/experiment/code), and academic-writing-skills (skills/prompts/tests/documents). Each repository remains one case, and the double-anonymous framing treats them only as author-associated local projects. The main corpus includes complete repository-direct Claude, Codex, and Gemini sessions and retains calls without resolved file effects.

Global path-only matches are a separate sensitivity source because they omit surrounding no-file calls.

The corpus comprises 551 native root sessions, deduplicated from 2,049 source transcript files, and 181,303 Tool actions; 550 roots and 175,619 actions resolve to retained worktrees. The repaired projection yields 5,792 artifact identities and 13,905 confirmed mutation rows. All six cases pass the longitudinal gate, but each estimand retains its own source-coverage denominator. Primary outputs are project-level effect sizes, distributions, cumulative-incidence curves, and explicit coverage. Model-derived motifs or summaries are exploratory unless they resolve to source IDs and deterministic metrics.

**Admission gates.** The longitudinal gate requires admitted native roots plus worktree-attributed actions, artifacts, and mutations for a project-level case. Cross-case estimands proceed only when at least four projects have a defined denominator. RQ4 additionally requires 20 eligible boundaries within each contributing project, while an exact-context RQ5 Skill stratum requires at least three distinct native-root sessions. A stopped gate yields within-case or coverage evidence, never an imputed zero.

**RQ1—Persistence, reuse, and repeated mutation.** We ask whether observed introductions persist, whether later work returns to mutated lineages, and whether activity volume orders those outcomes. All six cases contain eligible confirmed introductions after projection repair and therefore support introduced-artifact persistence: the project proportions are 974/1043, 28/245, 18/30, 8/18, 18/18, and 1/1. Later reuse appears for 89.29–97.02% of eligible mutations. Its descriptive cross-project association with worktree-attributed action volume is only Spearman  $\rho = 0.2000$ , so action count weakly orders even this widely covered relation in the selected cases (Figure 1). Repeat-observed episodes account for 74.5–90.8% of mutation episodes, and the most-mutated 10% of identities account for 42.0–86.7%. These distributions establish recurring, concentrated work, not convergence or thrashing.

**RQ2—Validation response.** We ask how recognized successes and failures interleave with mutation bursts and worktree-local mutation accumulation. Projection repair raises recognized-success coverage from three to all six projects, so the predeclared four-project gate now admits a cross-case description. Across seven eligible worktree lanes, complete intervals between recognized successes are strongly zero-inflated: 29.3–86.8% contain no mutation, while lane maxima range from 1 to 817 mutations. The corresponding per-project validation-before-supersession proportions are 2424/6111, 1212/5604, 141/694, 47/282, 9/196, and 33/247. This measures recognized cadence, not redundant testing or validation coverage of each mutation. Mutation effects project to the affected worktree, whereas a validation stays on the command worktree; incomplete intervals at observation end are not silently counted as complete. RQ2 therefore finds that many adjacent recognized successes contain no intervening mutation, yet rare intervals accumulate long mutation bursts.

**RQ3—Workspace focus evolution.** We ask how path-resolved access and mutation are allocated and how artifact

and module focus changes in native call order. Path-resolved calls are analyzed in native order as same-artifact, same-module, or cross-module transitions, with access and mutation and `ok` and `observed` statuses kept distinct. All six cases now carry return evidence after the native-root session-join repair, and pooled unknown-create births fall from 790 to zero. Status sensitivity can be material: in one case, mutation allocation to `paper/docs` is 39.2% over all resolved statuses but 88.2% for `ok` only, so neither view is silently promoted to ground truth. In the path-compatible local anchors used by RQ6, cross-module movement is 2.1–20.2%, so 79.8–97.9% of adjacent transitions stay on the same artifact or within its top-level module. Qualified local module returns have medians of 2–4 strictly intervening calls. RQ3 therefore finds predominantly path-local movement with short observed returns. These path relations do not measure duration, internal attention, importance, productivity, cooling, or forgetting.

**RQ4—Cross-session continuity.** We ask what source-verifiable re-grounding and workspace continuity occurs between independent, temporally ordered session components. Streams sharing a native root remain one independent block, and transitive concurrency components prevent us from inventing order among overlapping sessions. Only adjacent non-overlapping components are compared for pre-mutation re-grounding and prior artifact/module overlap. The projection forms 121 components and 108 boundaries, but timing is defined only when a first mutation is observed and empty denominators remain undefined. After corrected root/subagent joining, only three of six projects reach 20 eligible boundaries; the preregistered four-project estimator gate therefore still stops. RQ4 is a data-limited within-case and coverage result, not evidence of reset cost, memory, comprehension, or forgetting.

**RQ5—Skill and instruction footprints.** We ask whether source-attributed named-Skill action composition is repeatable within exact contexts, without turning association into a harness effect. A separate checker reconciles all 2,063 included native transcript streams, including 7,304 Skill/instruction signal rows and 205,836 adjacent Tool boundaries. The cases contain 67 explicit Skill invocations and 1,675 source-attributed Skill Tool actions. Invocation and execution do not always share one transcript stream: only 476 attributed actions are contiguously preceded by a matching same-stream invocation, which is a coverage diagnostic rather than an inferred delegated episode. Five exact-context named-Skill strata qualify, but only one project supports a two-Skill comparison. There, median Jensen–Shannon distance is 0.116 for nine same-Skill pairs and 0.123 for ten different-Skill pairs; the one-sided exact root-block test gives  $p = 0.750$  over 12 admissible assignments. Thus the only eligible comparison supports no repeatable Skill fingerprint. Accesses to `AGENTS.md`, `CLAUDE.md`, and `SKILL.md` remain separate focal events and do not prove injection, compliance, utility, or harm.

**RQ6—External boundary.** We ask whether compatible path-local relations recur outside the selected long-running cases and explicitly stop where public sources lack lineage.

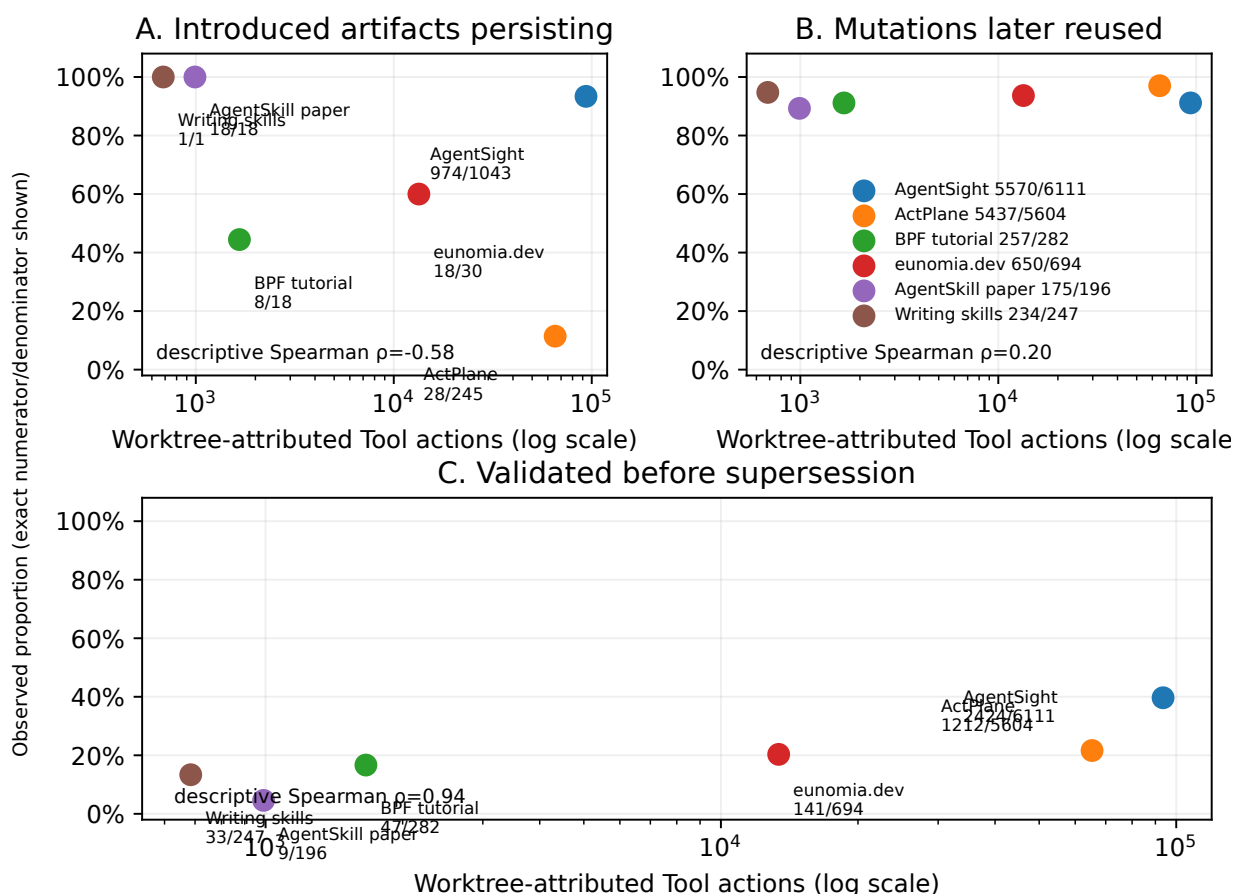


Figure 1: Activity volume versus the three separately reported RQ1 proportions. All dimensions cover six cases after repair; labels show exact numerators/denominators. Correlations are descriptive across six projects, not population estimates.

We select 64 independent units within each of four Open-SWE-Traces harness/model strata and within the IdeaTrail topic stratum: 320 stratum-specific selections. Across strata, the 256 Open-SWE selections contain 255 unique task IDs, so independence is not claimed globally. An independent parser reconciles 31,249 Tool calls and 22,113 adjacent path-target transitions. Cross-module movement is 18.0–30.0% publicly versus 2.1–20.2% in compatible local anchors, and module-return medians are 2–3 intervening calls publicly versus 2–4 locally. Path locality and short returns recur, with a magnitude difference; persistent artifact lineage, cross-session re-grounding, and exact Skill attribution are unavailable and remain N/A. Exploration-before-mutation is non-confirmatory because public harnesses shape action order, and recognized validation lacks a portable attempt/status pair. Path concentration and staged revision are analogies, not stable-artifact replication.

The supplement reports the full gate definitions, coverage tables, per-project denominators and curves, status sensitivities, checker procedures, and the nine empirical figures omitted here.

## Measurement-Capability Conformance

We freeze 12 complete native files per project and derive 120 questions: 30 each about action-only (A), artifact-linked (B), cross-session (C), and final-state (D) facts. A standalone checker imports neither the projection nor `agent-session`; it reparses all 72 files, reconstructs 1,721 artifact edges, and verifies cutoff state.

**Frozen failure.** On 2026-07-23, the trajectory answered all 60 B+C questions but only 32 correctly, with per-project conditional accuracy ranging from 0.0 to 1.0. Final State answered all 30 D questions, while ProcGrep and Trajectory produced identical answers on all 30 A questions and matched the frozen source-direct grammar on 18; the official action adapter deliberately leaves some terminal reads unclassified.

**Taxonomy.** Of the 28 disagreements, 14 arose from deliberately broader shell/scope evidence and 14 from genuine bugs (seven native-root joins, four failed-call effects, and three shell-path extractions); the audit also exposed an event-workdir bug. The independent oracle was corrected to v4, changing 24/120 expected answers with question-level source justification.

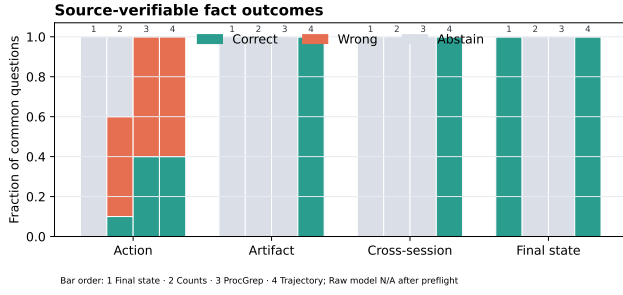


Figure 2: Correct, wrong, and abstaining outcomes on the corrected 120-question oracle. Raw is N/A after a retrieval-engaged preflight boundary stop; it is not scored as wrong or abstaining.

**Repaired conformance.** After fixing all four root-cause classes, the current revision scores B 30/30, C 30/30, and D 30/30 against v4 (Figure 2). Its A-family score is 12/30 because the trajectory deliberately preserves ProcGrep’s action spine while v4 re-derives atoms under a source-direct grammar. The result is 60/60 B+C repair-corpus conformance, not a general exact-fact capability claim.

Final State, Counts, and pinned ProcGrep remain lower-information controls. The bounded Raw reader is N/A: its only retrieval-engaged preflight stopped after 11 local evidence calls and 117,184 retrieved bytes, when evidence containing an original absolute path crossed the frozen harness boundary. Thus none of 360 planned Raw rows ran. This is a reader–harness contract incompatibility rather than model performance; we make no Raw accuracy, cost, efficiency, or superiority claim. Deterministic timing is also not method-specific because the four completed methods share one project-loop measurement.

## Threats to Validity and Ethics

Native records may omit subprocess effects, external edits, and uninstrumented actions; missing fields are coverage gaps, not no-effect observations. Final existence cannot prove content survival, and recognized success is association rather than coverage or correctness. Optional system-level evidence is a future coverage extension, not a second truth layer in this study. Existing-file writes retain unknown content durability unless native diffs, snapshots, or Git evidence support it independently.

The conformance audit found material disagreement rather than assuming the projection correct. Genuine path, identity, root-join, failed-effect, and workdir errors were repaired; RQ1–RQ4 were then recomputed. The repaired 60/60 B+C result is limited to this corpus and corrected oracle, and the action-only grammar difference remains explicit. RQ5 and RQ6 are independently reconstructed, but Skill, model, task, phase, and project remain confounded. Repeated actions, unvisited artifacts, and long validation gaps are observable process facts, not automatic failure labels.

The six cases are selected from local author-associated projects. They offer deep source-native histories but cannot estimate population rates; public Open-SWE-Traces attempts

are synthetic, and IdeaTrail is reverse-synthesized under one workflow. They replicate only compatible within-attempt relations and cannot validate natural cross-session persistence, re-grounding, or Skill effects. Native trajectories can contain private prompts, code, paths, secrets, and experimental data. Release requires repository and data permission, credential and personal-content removal, and aggregate or explicitly approved examples. Process measures must not become human-productivity scores: we do not rank people, equate activity with progress, or label repeated work as waste without independent task evidence. The intended use is analysis by an Agent or automated diagnostic process over its own authorized workspace.

## Related Work

Software-agent trajectory studies measure actions, testing, and strategy (Bouzenia and Pradel 2025; Mehtiyev and Asunção 2026); TRAJEVAL identifies coherence collapse after useful intermediate code (Kim et al. 2026); and ProcGrep provides action procedures and fingerprints (Oderinwale 2026). AgingBench and plan-persistence work study long-session reliability and context (Zhu et al. 2026; Mehta and Datta 2026), while OR-Space supplies persistent multi-artifact tasks (Zhou et al. 2026). AgentDiagnose and AgentRx diagnose completed traces (Ou et al. 2025; Barke et al. 2026); TrajAudit retrieves trace evidence (Wang, Xie, and Huo 2026); and HarnessFix links failures to harness mechanisms (Chen et al. 2026). We do not claim the first action analysis, fingerprint, persistent workspace, or diagnostic labeler. Our contribution is source-linked artifact identity and cross-session lineage for measuring real project histories.

## Conclusion

Persistent workspaces, not isolated sessions, expose whether long-running Agent activity survives, receives later reuse and recognized validation, concentrates in repeatedly changed artifacts, and continues across session boundaries. Six cases show high later reuse but weak ordering by action volume, alongside hard coverage limits for continuity and Skills. Public traces reproduce path locality and short module returns but not longitudinal lineage. Finally, a frozen 32/60 B+C conformance failure led to root-cause repair and 60/60 on the corrected corpus oracle, without implying a general exact-fact capability.

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